

## Practice sheet: Natural and Step Responses of RL and RC Circuits

This sheet is not marked for correctness, and AI tools are allowed. The device-free quiz is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Get both right; on the quiz each earns half the marks.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Conventions:** “a long time” means the circuit has reached dc steady state: an inductor is a short circuit and a capacitor an open circuit. Every response here has the form  $x(t) = x_f + (x_0 - x_f)\exp(-t/\tau)$ , with  $\tau = L/R$  or  $\tau = RC$ , where  $R$  is the resistance the inductor or capacitor sees after switching. Check every answer:  $x(0)$  must equal the initial value,  $x(\infty)$  the final value, and  $\tau$  must be positive.

**S1. What cannot jump? (two-tier)** A switch moves at  $t = 0$  in a circuit with a resistor and an inductor. Which of these must always have the same value just after the switch moves ( $t = 0^+$ ) as just before ( $t = 0^-$ ), in any circuit of this kind?

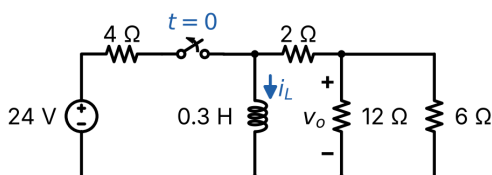
- |                           |                           |
|---------------------------|---------------------------|
| a) The inductor's voltage | b) The inductor's current |
| c) The resistor's current | d) The resistor's voltage |

*Reason:*

- An inductor's current cannot change instantly: a jump would need an infinite voltage.
- An inductor's voltage cannot change instantly, because its current is continuous.
- A resistor's current cannot jump, because Ohm's law fixes it.
- A resistor's voltage cannot jump, because the resistance does not change.

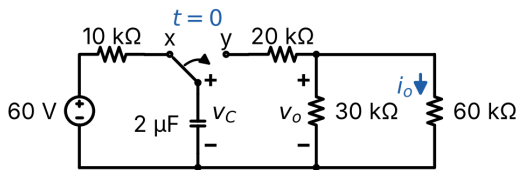
*Follow-up:* name the quantity that cannot jump in a circuit with a capacitor.

**S2. RL natural response.** As in Nilsson Example 7.1. The switch has been closed for a long time and opens at  $t = 0$ .



- Find  $i_L(0)$ .
- Find the time constant  $\tau$ .
- Find  $i_L(t)$  for  $t \geq 0$  and  $v_o(t)$  for  $t \geq 0^+$ .
- Find  $v_o(0^-)$  and  $v_o(0^+)$ . Why can  $v_o$  jump when  $i_L$  cannot?
- What percentage of the initial stored energy is dissipated in the  $12\ \Omega$  resistor?

**S3. RC natural response.** As in Nilsson Example 7.3. The switch has been in position x for a long time and moves to position y at  $t = 0$ .



- Find  $v_C(0)$ .
- Find  $\tau$ .
- Find  $v_C(t)$  for  $t \geq 0$ , and  $v_o(t)$  and the current  $i_o$  down through the  $60 \text{ k}\Omega$  for  $t \geq 0^+$ .
- How much energy is dissipated in the  $60 \text{ k}\Omega$  resistor?

**S4. One time constant later (two-tier).** In a natural response, what fraction of the initial value remains after one time constant?

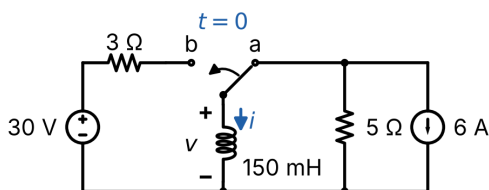
- 63%
- 0
- 37%
- 50%

*Reason:*

- The time constant is the time the response takes to reach its final value.
- After one time constant,  $x = x_0 e^{-1} = 0.37x_0$ .
- Half of what remains decays in each time constant.
- After one time constant,  $1 - e^{-1} = 63\%$  of  $x_0$  remains.

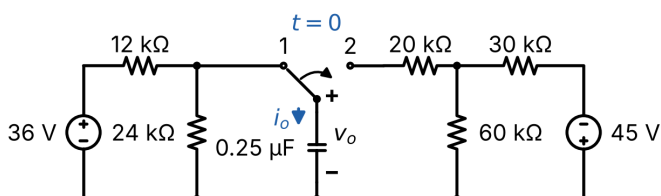
*Follow-up:* after how many time constants is less than 1% left?

**S5. RL step response.** As in Nilsson Example 7.5. The switch has been in position a for a long time. At  $t = 0$  it moves to position b; it makes contact at b before it breaks contact at a, so the inductor current is never interrupted.



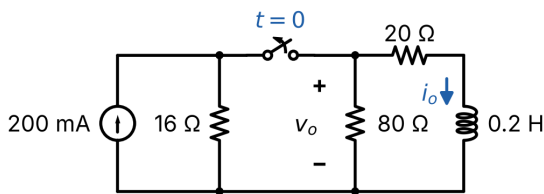
- Find  $i(0)$ ,  $i(\infty)$  and  $\tau$ .
- Find  $i(t)$  for  $t \geq 0$ .
- Find  $v(t)$  for  $t \geq 0^+$ . Why is  $v(0^+)$  larger than the 30 V of the source?
- When does  $v$  equal 30 V?

**S6. RC step response with a Thévenin equivalent.** As in Nilsson Example 7.6. The switch has been in position 1 for a long time and moves to position 2 at  $t = 0$ .



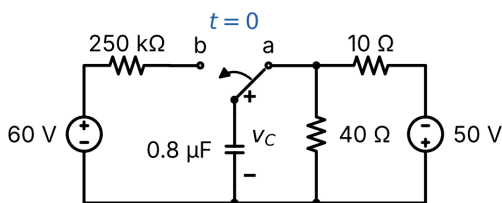
- Find  $v_o(0)$ .
- For  $t \geq 0$ , find the Thévenin equivalent that the capacitor sees.
- Find  $v_o(t)$  for  $t \geq 0$ .
- Find the capacitor current  $i_o$ , referenced into its + terminal, for  $t \geq 0^+$ .

**S7. The general method, RL.** As in Nilsson Example 7.7. The switch has been closed for a long time and opens at  $t = 0$ .



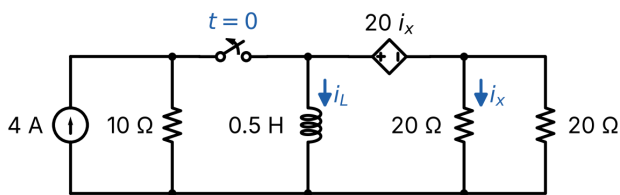
- Find  $i_o(0)$ .
- Find  $\tau$ .
- Find  $i_o(t)$  for  $t \geq 0$ .
- Find  $v_o(0^-)$ ,  $v_o(0^+)$  and  $v_o(t)$  for  $t \geq 0^+$ .

**S8. The general method, RC from a negative voltage.** As in Nilsson Example 7.8. The switch has been in position a for a long time and moves to position b at  $t = 0$ .



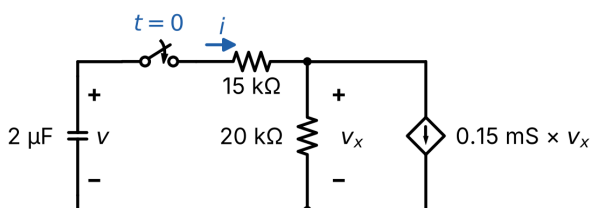
- Find  $v_C(0)$ . Watch the polarity.
- Find  $v_C(t)$  for  $t \geq 0$ .
- Find the capacitor current  $i$ , referenced into its + terminal, for  $t \geq 0^+$ .
- When is  $v_C$  zero?

**S9. A dependent source sets  $\tau$ .** As in booklet Problem 7.7. The switch has been closed for a long time and opens at  $t = 0$ .



- Find  $i_L(0)$ .
- For  $t \geq 0$ , find the resistance the inductor sees, using a test source.
- Find  $\tau$  and  $i_L(t)$  for  $t \geq 0$ .
- Find  $i_x(t)$  for  $t \geq 0^+$ .

**S10. A dependent source in an RC circuit.** The  $2 \mu\text{F}$  capacitor has been charged to 30 V. At  $t = 0$  the switch closes.



- Find the resistance the capacitor sees, using a test source.
- Find  $\tau$  and  $v(t)$  for  $t \geq 0$ .
- Find  $i(t)$  and  $v_x(t)$  for  $t \geq 0^+$ .

**S11. Spot the error.** The worked solution below contains the kind of polarity slip AI chat tools often make. For  $t < 0$  a  $0.4 \mu\text{F}$  capacitor has been connected for a long time across the  $24 \text{ k}\Omega$  resistor of a divider: a  $36 \text{ V}$  source in series with  $12 \text{ k}\Omega$  and  $24 \text{ k}\Omega$ . The source's  $+$  terminal connects to the bottom of the  $24 \text{ k}\Omega$ , so the top of the  $24 \text{ k}\Omega$  is negative. At  $t = 0$  the capacitor is switched to a  $48 \text{ V}$  source in series with  $125 \text{ k}\Omega$ .  $v_C$  is positive at the top.

1.  $\tau = (125 \text{ k}\Omega)(0.4 \mu\text{F}) = 50 \text{ ms}$ .
2.  $v_C(0) = 36 \times 24/(12 + 24) = 24 \text{ V}$ .
3.  $v_C(\infty) = 48 \text{ V}$ .
4.  $v_C(t) = 48 + (24 - 48)e^{-20t} = 48 - 24e^{-20t} \text{ V}$ .
5. Check:  $v_C(0) = 24 \text{ V}$  and  $v_C(\infty) = 48 \text{ V}$ , as found.

- a) Which is the first wrong line?
- b) Correct it and find  $v_C(t)$ .
- c) Line 5 calls itself a check. Why did it not catch the error?
- d) When is  $v_C$  zero?

**S12. Spot the error in a time constant.** An uncharged  $0.4 \mu\text{F}$  capacitor is connected at  $t = 0$  through  $5 \text{ k}\Omega$  to a node N. Node N connects through  $30 \text{ k}\Omega$  to a  $60 \text{ V}$  source and through  $60 \text{ k}\Omega$  to ground.

1.  $v(0) = 0$ , since the capacitor is uncharged.
2.  $v(\infty) = 60 \times 60/(30 + 60) = 40 \text{ V}$ .
3.  $\tau = (5 + 30 + 60 \text{ k}\Omega)(0.4 \mu\text{F}) = 38 \text{ ms}$ .
4.  $v(t) = 40 - 40 \exp(-t/0.038) \text{ V}$ .

- a) Which is the first wrong line?
- b) Find the correct  $\tau$  and  $v(t)$ .
- c) What resistance does the capacitor see, and how do you find it?

**S13. Work backwards.** A capacitor charges through  $10 \text{ k}\Omega$ . Its voltage is  $10 \text{ V}$  at  $t = 0$ , rises toward  $40 \text{ V}$ , and reads  $30 \text{ V}$  at  $t = 2 \text{ ms}$ .

- a) Find  $\tau$ .
- b) Find  $C$ .
- c) When does the voltage reach  $39 \text{ V}$ ?
- d) The same steps work for an inductor: a current rises from  $0$  toward  $2 \text{ A}$  through  $20 \Omega$  and reads  $1.5 \text{ A}$  at  $t = 6.93 \text{ ms}$ . Find  $\tau$  and  $L$ .

**S14. An artificial pacemaker.** As in Nilsson's Practical Perspective for this chapter. A source  $V_s$  charges a capacitor  $C$  through a resistor  $R$ , starting from  $0 \text{ V}$ . When  $v_C$  reaches  $0.75V_s$ , a controller discharges the capacitor in a negligible time and sends a pulse to the heart; then charging starts again.

- a) Show that each charging takes  $t = RC \ln 4$ , so the heart rate is  $H = 60/(RC \ln 4)$  beats per minute.
- b) With  $C = 2 \mu\text{F}$ , what  $R$  gives 60 beats per minute?
- c) With  $C = 2 \mu\text{F}$  and  $R = 300 \text{ k}\Omega$ , what is the heart rate?
- d) *Predict first:* what happens to the heart rate if  $R$  is doubled?

**S15. How fast does a relay let go?** A 0.5 H relay coil with 60  $\Omega$  winding resistance carries 0.2 A from a 12 V supply. When the driving switch opens at  $t = 0$ , the coil current continues through a diode (ideal: 0 V when conducting) in series with a 140  $\Omega$  resistor, connected across the coil.

- Find  $\tau$  for  $t \geq 0$ .
- The relay releases when the current falls below 50 mA. When does it release?
- What voltage appears across the coil's terminals at  $t = 0^+$ ? Take it positive at the terminal the supply made positive.
- Repeat (a) and (b) with the diode alone (no 140  $\Omega$ ). Why do designers sometimes add the resistor?

**S16. Check an answer without solving again.** A 0.4 H inductor carries  $i(0) = -2$  A. At  $t = 0$  it is switched across a 20 V source in series with 5  $\Omega$ ;  $i$  is referenced in the direction the source drives current. Student 1:  $i = 4 - 6 \exp(-12.5t)$  A. Student 2:  $i = 4 - 2 \exp(-12.5t)$  A. Use  $i(0)$ ,  $i(\infty)$  and  $\tau$  to decide which student is right.